

11-Python String Methods and Conditional Statements

Python String Methods

- Introduction and practical use of common Python string methods, focusing on those frequently used in coding assignments.

- **String Formatting:**

- f-string (available from Python 3.5+): `f"A={A}, B={B}, C={C}"`

- `.format()` method: `"A={}, B={}, C={}".format(A, B, C)`

- **Startswith/Endswith:**

- `str.startswith(substring, start, stop)`

- : Returns True if string starts with substring in given range.

- `str.endswith(substring, start, stop)`

- : Returns True if string ends with substring within range.

- Start index is inclusive; stop index is exclusive, similar to slicing.

- **Find/rfind and Index/rindex:**

- `find(substring, start, stop)`

- : Returns index of first occurrence (left to right) or -1 if not found.

- `rfind(substring)`: Returns index of last occurrence (right to left) or -1 if not found.

- `index(substring)`: Like `find()`, but raises `ValueError` if substring not found.

- `rindex(substring)`: Like `rfind()`, but raises `ValueError` if not found.

- **Count:**

- `count(substring, start, stop)`
: Counts occurrences of substring within string or specified range.

- **String Alignment & Filling:**

- `center(width, fillchar)`: Centers string, pads with fillchar (default: space).
- `ljust(width, fillchar)`: Left-justifies, pads right.
- `rjust(width, fillchar)`: Right-justifies, pads left.
- `zfill(width)`: Zero-fills to right-justify numbers as strings (only takes numeric strings).

- **Whitespace & Character Stripping:**

- `strip(chars)`: Removes leading/trailing spaces or specified characters.
- `lstrip(chars)`: Removes from left.
- `rstrip(chars)`: Removes from right.
- Does not remove anything in the middle of the string.

- **Replace & Split/Join:**

- `replace(old, new, count)`
: Replaces occurrences of old with new (optionally limited by count).
- `split(sep, maxsplit)`: Splits string into list; separator default is space; maxsplit limits splits.
- `join(iterable)`: Joins list/tuple of strings into a single string using specified separator.

- **Case Handling:**

- `lower()`: Converts ASCII chars to lowercase.
- `casefold()`: More aggressive; handles unicode and converts special cases (e.g., German ß to ss).

- **Chaining Methods:**

- Possible if intermediate return values are strings; e.g., combine strip and lstrip.

- **Methods Not Covered in Detail:**

- `format_map`, `expandtabs`, `partition`, `splitlines`, `translate`, `maketrans`, `encode` – students encouraged to check documentation.

Conditional Statements

- **Keywords:** `if`, `elif`, `else`.

- **Basic Structures:**

- Single `if`: Checks one condition.
- `if-else`: Chooses between two blocks.
- `if-elif-else`: For multiple mutually exclusive conditions.

- **Examples:**

- Odd/Even checker: Checks number parity using modulus, prints result.

- Weekday selector: Uses `if-elif-else` to print day corresponding to input number (1-7); else catches invalid inputs.
- Leap year checker (nested if):
 - Checks divisibility by 4, then by 100, and finally by 400 for century years.
 - Logic: Divisible by 4? If so, is it a century (divisible by 100)? If so, must also be divisible by 400 to be leap year.
- Discussion of alternate logical operator implementation (using and/or in a single if condition).
- **Coding Practices:**
 - Importance of indentation to define code blocks.
 - `f` before string for output formatting and the use of `.format()`.
 - Using logical operators for combining multiple conditions in one `if` statement.

Assignments & Submission

- String and conditional statement assignments are provided; string assignment due on 17th, both string and loop assignments due on 28th.
- Assignments to be submitted via email (address provided in class); each topic in a separate file.
- In assignment answers, students should include explanations for truth table and reasoning behind expected errors/outputs.

Interaction & Q&A

- Clarifications on methods, logic behind code, differences between methods, and detailed walk-throughs of sample problems.
 - Reminders that string methods return new strings (immutability), and certain operations (like chaining) only possible when the returned value is string type.
 - Guidance on Boolean assignment specifics, object reusability, and logic explanations for exceptional outputs.
 - Students encouraged to copy code into Python editor or online tables to understand execution order.
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Possible Detail Questions Based on Summary:

- Which string methods handle case sensitivity? How do find/rfind differ from index/rindex?
- How to split and join strings in Python?
- What are the rules for leap year calculation?
- What format is expected for assignment submission?
- How does strip differ from replace for whitespace?
- What are the best practices for indentation in Python?
- How do logical operators affect conditional execution?
- What to do if output is an error or exception in assignment answers?

